PORTFOLIO PROJECT PDF

Automatidata Portfolio Predicting Generous Tippers

This is my updated portfolio version of the Automatidata project from the Google Advanced Data Analytics program. I used official NYC Open Data, rebuilt the workflow in Colab, and compared baseline and tree-based models to predict whether a credit-card taxi trip ended with a tip rate of 20 percent or higher.

Python | Colab | NYC Open Data API | Classification | XGBoost | Random Forest

PROJECT INFO

At a glance

Course reference: Google ADA - Automatidata

Project type: Individual portfolio rewrite

Dataset: 2017 Yellow Taxi Trip Data

Raw slice pulled: 180,000 rows

Modeling sample: 116,748 trips

Rows kept: Credit-card trips only

Primary metric: F1 score

Champion model: XGBoost

Important note: selected by a narrow validation margin over Random Forest

BUSINESS PROBLEM

The original scenario focused on non-tippers, but that can create unfair access issues. In this version, I reframed the task into predicting generous tippers so the project stays useful for revenue analysis while being more responsible.

WHY THIS VERSION WORKS FOR GITHUB

I replaced the course CSV setup with direct public-data download, added a cleaner train / validation / test flow, and kept the writing simple so an interviewer can understand the project quickly.

TEST F1

76.5%

Balanced summary of precision and recall for the final model.

TEST RECALL

84.6%

Strong ability to capture actual generous tippers.

TEST ACCURACY

72.7%

Good overall classification accuracy on the holdout set.

POSITIVE CLASS

52.7%

The target is close to balanced, so F1 is suitable here.

PROJECT OVERVIEW

Business Problem, Proposal, and Stakeholders

I wanted this PDF to work like a quick project brief. The goal is to show the business problem, the data workflow, and the main result without making the reader go through a long technical notebook first.

BUSINESS FOCUS

Tip behavior

Estimate which trips are more likely to end with a 20 percent or higher tip.

PROJECT PROPOSAL

Responsible target

Use generous tipping instead of non-tipping to reduce ethical risk in the original scenario.

STAKEHOLDERS

TLC + drivers

Useful for a high-level decision support idea, not as a strict dispatch rule.

STAR-STYLE PROJECT STORY

Situation: the Automatidata scenario asked for a tip-related model for NYC taxi trips.

Task: rebuild the project as a public portfolio version with official open data and cleaner modeling logic.

Action: downloaded data by API, filtered to credit-card trips, engineered behavior and time features, and compared three classifiers.

Result: XGBoost reached 0.765 F1 on the test set and gave a clear, reusable project story for GitHub.

Section	What I kept or changed
Course idea	Kept the Automatidata business theme and PACE structure.
Data access	Changed to direct NYC Open Data API instead of local course CSV files.
Target	Kept generous tipper logic: tip rate >= 20 percent.
Model flow	Added train / validation / test split for cleaner final selection.
Interpretation	Added permutation importance for an easier portfolio explanation.

MAIN QUESTIONS

- Can public trip-level data predict generous tipping with usable signal?
- Which trip or fare features matter most in this portfolio version?
- How close are the tree-based models on validation and test data?
- What limitations should be stated clearly before showing this result?

SIMPLE WORKFLOW

API download -> clean -> keep credit-card trips -> engineer features -> split data -> train baseline and tree models -> compare by F1 -> explain limits and next steps.

> **Portfolio note**
> This project is designed for clear communication. The notebook contains more detail, but this report keeps only the most useful parts for interview reading.

DATA AND METHODS

Dataset, Feature Engineering, and Selected Code

The dataset came from official NYC Open Data. I pulled a manageable public slice, then narrowed the modeling set to trips where tip values are more reliable.

RAW PULLED
180k
Monthly API chunks from the 2017 table.

AFTER FILTERING
116,748
Valid credit-card trips used for modeling.

RAW COLUMNS
13
Trip fields from the API slice before feature engineering.

MODEL INPUTS
21
Final modeling columns before one-hot encoding in the pipeline.

Dataset summary	Details
Main table	2017 Yellow Taxi Trip Data from NYC Open Data.
Pull method	CSV endpoint through the Socrata API in monthly chunks.
Filtering logic	Credit-card trips only, positive fare, positive distance, valid durations, and reasonable passenger counts.
Target	Generous = 1 when tip_percent >= 0.20.
Important limit	The final model represents the filtered credit-card subset, not all taxi riders.

FEATURE EXAMPLES

Time: pickup hour, day, month, rush-hour flags

Trip: distance, duration, speed, passenger count

Fare behavior: fare amount, fare bins, round5_ratio, round10_ratio

Location: grouped pickup and dropoff zones

MODELS COMPARED

Logistic Regression, Random Forest, and XGBoost. Final model choice was based on validation F1, then tested once on the holdout set.

SELECTED CODE - OFFICIAL DATA PULL

```
API_ENDPOINT = "https://data.cityofnewyork.us/resource/
biws-g3hs.csv"
selected_columns = [
    "vendorid", "tpep_pickup_datetime",
"tpep_dropoff_datetime",
    "passenger_count", "trip_distance", "ratecodeid",
    "pulocationid", "dolocationid", "payment_type",
    "fare_amount", "tip_amount", "tolls_amount",
"total_amount"
]

# pull monthly chunks, combine, clean, and model in
Colab
```

This changed the project from a course-only notebook into a public and reproducible portfolio workflow.

SELECTED CODE - TARGET AND ENGINEERED FEATURES

```
df["tip_percent"] = np.round(
    df["tip_amount"] /
(df["total_amount"] - df["tip_amount"]),
3
)
df["generous"] = (df["tip_percent"] >=
0.20).astype(int)

df["speed_mph"] = df["trip_distance"] /
(df["duration_minutes"] / 60)
df["round5_ratio"] =
((np.ceil(df["fare_amount"] / 5) * 5) -
df["fare_amount"]) / df["fare_amount"]
```

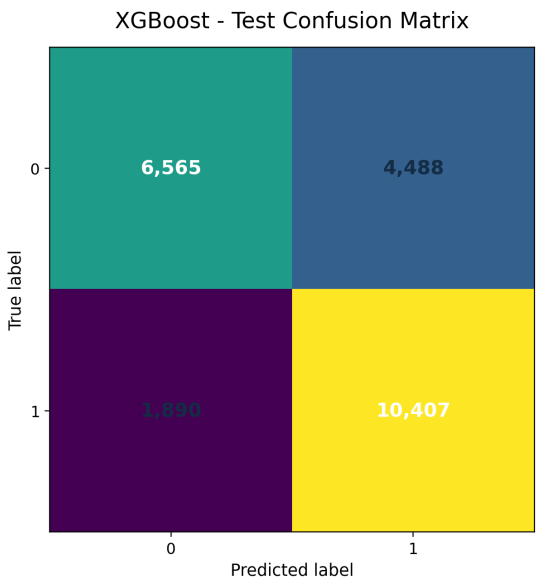
I kept the main idea from the course but updated the feature set to make the notebook more useful as a GitHub project.

RESULTS

Model Results and Visual Evidence

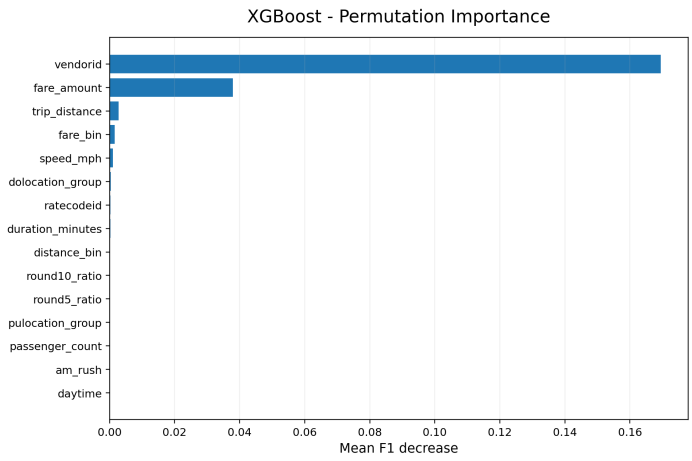
The charts below are the most important visual pieces in this report. I enlarged them so a reviewer can read the result quickly without going back to the notebook.

XGBoost - Test Confusion Matrix



Counts are shown as integers. The model captured many actual generous trips, but false positives are still meaningful and should be explained clearly.

XGBoost - Permutation Importance



Vendor and fare-related variables dominate the model. That is useful for interpretation, but it does not mean direct causality.

VALIDATION F1

76.58%

XGBoost finished just above Random Forest on validation.

TEST F1

76.54%

Very close to validation, so the result looks stable.

PRECISION

69.87%

The model still overpredicts some generous cases.

RECALL

84.63%

It captures most actual generous tippers in the test set.

WHAT THESE CHARTS TOLD ME

- XGBoost and Random Forest were almost tied, so the final choice should be described as a narrow win.
- The test result stayed close to validation, which supports the stability of this setup.
- VendorID is much stronger than the other features, so it may act as a proxy for rider or trip pattern differences.
- This model is useful for portfolio demonstration and insight work, but it should not be presented as a production dispatch rule.

Quick business read

The model shows that public trip data has usable signal for generous tipping, but the signal is uneven. It works better as an analysis prototype than as a high-stakes decision tool.

EXECUTIVE SUMMARY

Conclusion, Recommendations, and Interview Use

This project shows that I can take a guided classroom scenario, rebuild it into a public portfolio version, and explain the result in a business-friendly way.

MAIN FINDINGS

The filtered dataset was close to balanced, so F1 was a good main metric.

XGBoost performed best on validation and stayed consistent on test.

Fare and vendor features carried most of the predictive signal.

RESULT MEANING

The project shows that open trip-level data can support tip-behavior analysis, but the result should be framed as a prototype and not as a strict operational rule.

RECOMMENDATIONS

Collect rider-level history, improve location context, and test whether vendor effects stay strong after deeper statistical checks.

WHAT THIS PROJECT SHOWS ABOUT MY WORK

- I can turn a course lab into a cleaner public portfolio project.
- I can use official public data and make the workflow reproducible.
- I can compare baseline and tree-based models instead of showing only one result.
- I can explain model strengths and limits in simple English.

HOW I WOULD EXPLAIN THIS IN AN INTERVIEW

Simple version: I rebuilt the Automatidata classroom project using official NYC Open Data so it could run in Colab and be shared on GitHub. I changed the target to generous tipping, engineered time and fare features, compared three models, and selected XGBoost because it had the best validation F1 by a very small margin. The result was useful, but I would still present it as a portfolio prototype with clear limits.

> **Final note**
> This PDF is meant to be the easy-to-read project report inside my GitHub portfolio folder. The full notebook and charts are still available for technical review.

One-page Executive Summary Preview

Machine Learning Model Outcomes

Executive summary report for the New York City Taxi and Limousine Commission
Portfolio version prepared by Cheng Liu

Overview This project uses official NYC Open Data to build a classification model that predicts whether a yellow taxi rider is a generous tipper on a credit-card trip.

Problem The original non-tipper framing raises fairness concerns. In this version, the target is reframed to generous tippers, defined as riders with a tip rate of 20% or higher.

Solution I rebuilt the workflow as a public portfolio notebook in Colab, added direct API data access, used a train / validation / test split, and compared Logistic Regression, Random Forest, and XGBoost.

Details

Behind the data

- The model was trained on a reproducible 2017 public-data slice and filtered to credit-card trips, because cash tips are not recorded reliably in the trip table.
- The final sample had 116,748 trips and a nearly balanced target: 52.7% generous vs 47.3% not generous.
- XGBoost slightly outperformed Random Forest on validation F1 and was selected as the final model by a narrow margin.
- The strongest signals came from vendorid and fare-related features, but these are predictive patterns, not proof of causality.

Results Summary

The final XGBoost model reached an F1 score of 0.765 on the test set with high recall. This makes it a useful portfolio example and a reasonable analytical prototype, but not a production decision system.

model	precision	recall	F1	accuracy
LogReg val	0.7056	0.7550	0.7295	0.7051
RF val	0.6964	0.8492	0.7652	0.7256
XGB val	0.6960	0.8511	0.7664	0.7258
XGB test	0.6987	0.8463	0.7654	0.7269

Selected model metrics from validation and test results

Future model suggestions

- Add customer history or rider-level behavior features if available.
- Test threshold tuning to reduce false positives for drivers.
- Check the vendorid effect with additional segmentation or statistical analysis.

Next Steps

This project is suitable for a public GitHub portfolio because it uses official public data, a reproducible Colab workflow, and a cleaner evaluation design than the course notebook. The model should still be presented as a learning-focused prototype with clear limits, especially because the false-positive count remains higher than the false-negative count.

I also prepared a separate one-page visual summary for faster screening. This works well as a companion asset in the GitHub repository.